

IIOT :- Assignment No: 02 Q10. + Question Bank U2

- ✓ 1) Explain the following layers in Core IOT functional Stack-
 - A) Gateways and Backhaul Sublayer
 - B) Communications Network Layer
- ✓ 2) Which IOT protocol is used for smart home automation? How will you provide security for these applications with suggested protocol?
- ✓ 3) Explain the concept of Edge computing, fog computing and cloud computing. How they are related to each other?
- ✓ 4) Explain at least five use cases where IOT involvement will convert cities into Smart Cities.
- ✓ 5) Compare with suitable factors Microsoft Azure IOT and Google Cloud IOT used as IOT software platforms.
- ✓ 6) Give the function of each layer of a seven-layer IOT architectural reference model published by IOTWF architectural committee.
- ✓ 7) Compare with suitable factors Particle Photon with ESP32 and used for IOT application development.
 - 8) Explain different Energy related IOTs Domain with example.
- ✓ 9) Why RESTful JSON is a popular choice for IOT applications?
- ✓ 10) What is Fog Computing? Give advantages and disadvantages of fog computing.
- ✓ 11) What is Edge Computing? Give advantage and disadvantages of Edge Computing.
- ✓ 12) Explain in detail about Smart services in IOT system.

16/4/26

* Answers for UTI ans

Q7 Compare suitable factors particle photon with ESP32 and used for IoT application development.

Factor	Particle Photon	ESP32
Manufacturer	Particle Industries	ESPRESSIF system.
Connectivity	Wifi only	Wifi + Bluetooth.
Cloud Integration	Built-in Particle Cloud (easy device mgmt)	Node-fault Cloud (Use AWS, Azure, Firebase)
Ease of use	Very beginner friendly	Slightly complex, better for professional developers
Power of Consumption	Moderate	Low Power
Cost	Expensive	Low Cost
Security	Built-in Cloud Security	Supports encryption but needs manual up.

Q8 Compare with suitable factors Microsoft Azure IoT and Google Cloud IoT used for IoT platforms.

Parameter/ Factor	Microsoft Azure IoT	Google Cloud IoT.
Platform Overview	Comprehensive IoT suite with multiple services.	IoT Services integrated with Google Cloud sys.
Platform strength	Strong for big companies (Enterprise integration)	Best for AI & ML.
Core services	Azure IoT Hub.	Cloud IoT core.
Scalability	High	Very High
Ease of use	User-friendly dashboard low code tools	More developer-oriented require non-configuration
Security	Strong Security enterprise-grade security	Strong Google managed Security.
Protocols Supported	MQTT, AMQP, HTTP	MQTT, HTTP.

Q7 Explain atleast five usecases where IOT involvement will convert cities into Smart Cities.

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- IOT transforms traditional cities into "Smart Cities" by improving efficiency: —

① Smart Traffic mgmt: —

How IOT is used:

- Sensors, cameras, and GPS track vehicle movement.
- Traffic congestion signals adjust automatically based on congestion.

Benefits:

- Reduce traffic jams
- Shorter travel time.

② Smart Parking Systems: —

How IOT is used:

- Sensors detect vacant parking space.
- Mobile apps guide drivers to available spots.

Benefit:

- Saves time searching for parking.

③ Smart Street Lighting: —

How IOT is used:

- Light equipped with motion sensors and light sensors.

Benefit:

- Save Energy
- Improve safety.

④ Smart Health Care: —

How IOT is used:

- Wearable devices monitor patient health.
- Data send to hospitals in real time.

Benefits:

- Faster emergency response.
- Reduced hospital visit.

⑤ Smart Waste Management: —

How IOT is used:

- Smartbins notify garbage trucks only when they are full.

Benefits:

- Saving fuel and time.

Q7 Give the function of each layer of a Seven-layer IOT architectural reference model published by IOTWC architectural committee.

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① Physical Devices & Control layer: —

- Integrate sensor and actuators that collect raw data.
- Perform basic control actions.

② Connectivity: —

- Handle the transmission of data between the devices and network.

③ Edge Computing: —

- Performs local processing to reduce the amount of data sent to the cloud.

④ Data Accumulation: —

- Store data for long term use and convert it into readable format.

⑤ Data Abstraction: —

- Re-format data so that different applications can use it.

⑥ Application: —

- The software app where users see the data (like a dashboard).

⑦ Collaboration & Process: —

- Using data to make business or trigger real-world actions.

Q Explain in detail about Smart Servers in IOT systems

- Smart Servers are the "brain" of the IOT system that provides value to user.
- Smart device servers are value-added services built on IOT infrastructure that:
 - Collect data from physical devices.
 - Process and analyze that data.
 - Automatically trigger actions or provide insights.

- Types of Servers: —

A) Smart Home Servers: —

Smart Lighting, Smart thermostats, Voice assistance.

Eg., Light auto switch off when no is in the room.

B) Smart Transportation Services: —

Smart traffic Control Systems, Smart parking System.

C) Smart City Services: —

Smart Street lighting, Waste Management System.

- Working: —

1. Sensors collect data.
2. Data is sent via network.
3. Cloud/edge process network data.
4. AI analyze patterns.
5. Systems taken action or sends alerts.

- Example: —

A smart AC detect temperature → sends data → System analyzes → Automatically adjust cooling.

Q What's Fog Computing? Give advantages and disadvantages of Fog Computing

→ Definition: —

- Fog Computing is a distributed computing model where data processing, storage and analysis happen closer to the IOT devices (at the network edge) instead of relying entirely on distant Cloud Servers.

It is a layer of processing between devices and cloud.

Advantages: —

- 1) Reduced latency — Process data near the source.
- 2) Bandwidth Saving — only filtered/imp data is sent to cloud.
- 3) Better Reliability — Works even if Internet is weak.

Disadvantages: —

- 1) Complex Architecture — More layers, Harder to design.
- 2) Security — Requires proper security mgmt.
- 3) Higher cost — Requires additional H/W. (Gateways, local servers)

Q7 What's Edge Computing? Give advantages & disadvantage.

Definition: —

- Edge Computing is Computing Model where data is processed directly at or very close to the source of data (IoT device) instead of sending it to distant Cloud Servers.
- Edge computing happens directly on device or server at the R.

Advantages: —

- 1) Instant response — No network delay at all.
- 2) Privacy — Data stays on the device and is never sent over the Internet.

Disadvantages: —

- 1) Battery Drain — Processing data on device uses a lot of power.
- 2) Low power — Edge devices (like small sensors) aren't powerful enough for complex math or analysis.

Q8 Why RESTful JSON is a popular choice for IoT applications?

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- REST [Representation State Transfer]
 A web-based communication style using HTTP.
- JSON [JavaScript Object Notation]
 A light-weight data format for exchanging data.
- Reasons for popularity: —

A) Light weight Data format —

- JSON is compact and less verbose than XML
- Uses very little text by JSON; it uses less data & power to send.

B) Easy to understand —

- Human readable format
- Simple - key value structure

C) Language Independent —

- Works with almost any programming language (python, Java, C++, etc).

D) Security Supports —

- Works with HTTPS (SSL/TLS encryption)
- Supports authentication methods (API keys)

E) Standardized —

- Most web services and clouds already use RESTful JSON, making integration easy.